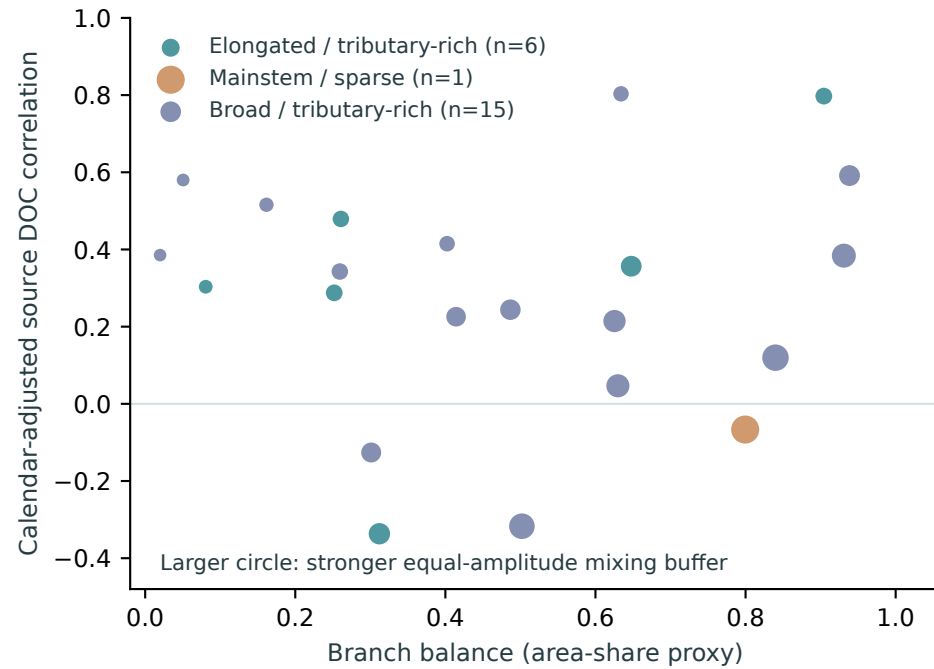
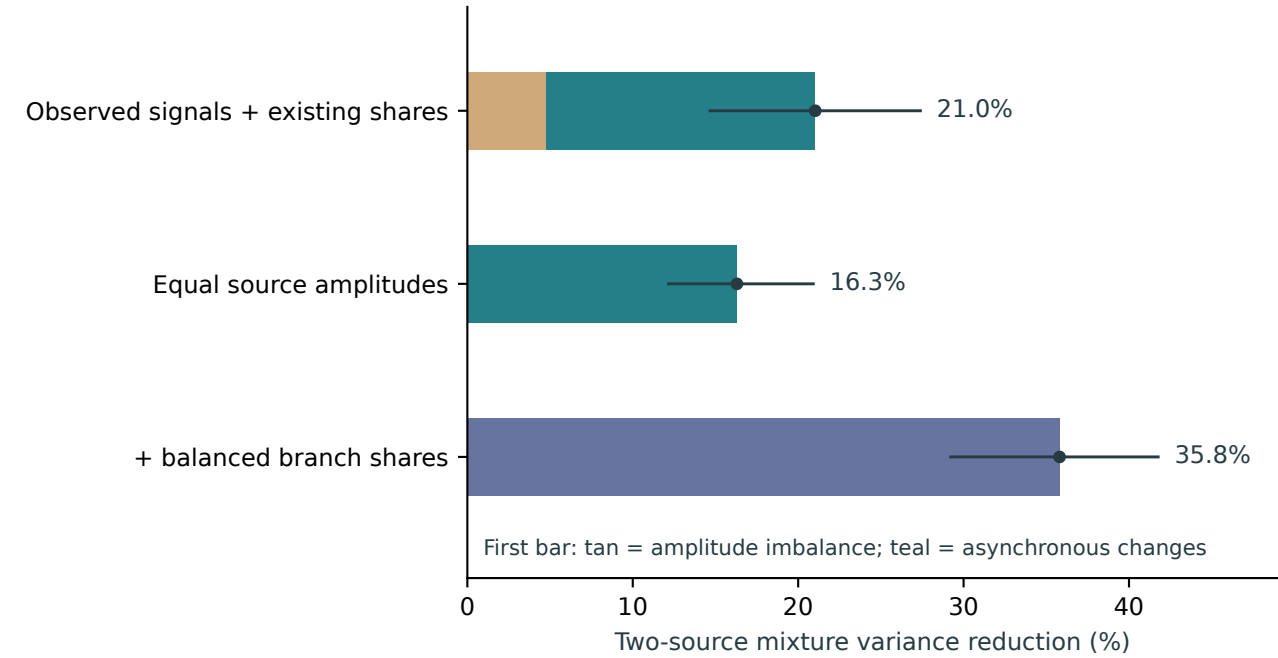


How can river-network operations modify DOC variability?

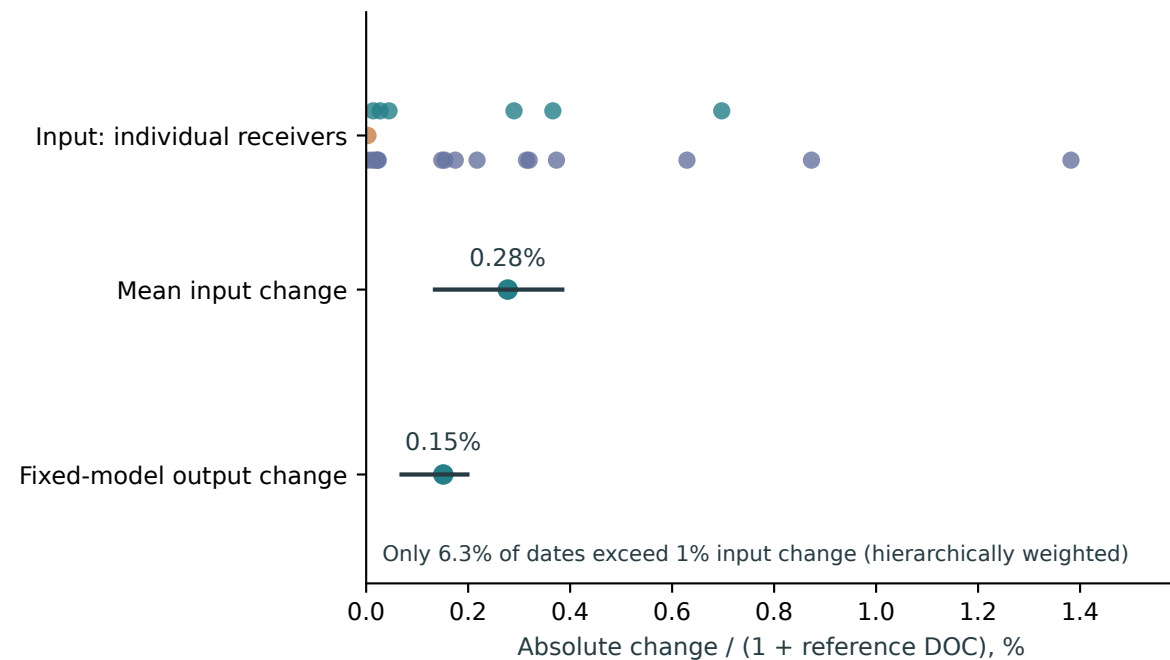
a Integration conditions in real tributaries



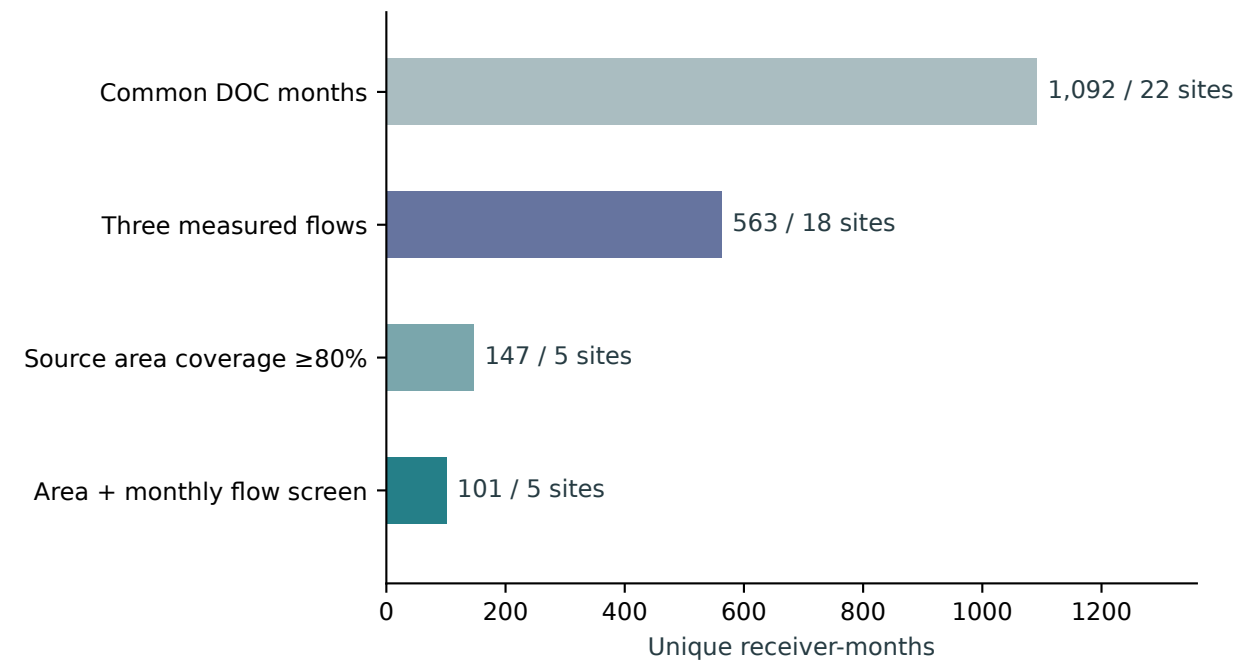
b Integration scenarios with source synchrony fixed



c Separate paths: monthly input sensitivity



d Channel loss requires a fuller water budget



59 real tributary pairs / 22 receivers / 11 monitoring systems; dates, pairs and receivers weighted hierarchically. Lines: system bootstrap 95%.

b: fixed mixing identity/scenarios, not whole-river removal. c: existing path interpolation, not measured velocity. Elongated/sparse forms lack independent systems.